AI-Based Email Classification and Priority Detection Using Transformers

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

DSA0306– Natural Language Programming

to the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE

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DECLARATION

We, **V. Pavan Kumar Reddy, G.Vinay Kumar Reddy** and of the **Artificial Intelligence &Machine Learning,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **'AI-Based Email Classification and Priority Detection Using Transformers'** is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**AI-Based Email Classification and Priority Detection Using Transformers**" has been carried out by **V. Pavan Kumar Reddy , G. Vinay Kumar Reddy** under the supervision of Dr. Kumaragurubaran T and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech **Artificial Intelligence** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Email has become an essential communication medium for individuals and organizations, resulting in a large volume of messages that users must manage every day. Identifying important and urgent emails manually can be time-consuming and may lead to critical messages being overlooked. This project proposes an AI-Based Email Classification and Priority Detection System Using Transformers to automatically categorize emails and determine their priority levels.

The proposed system uses Natural Language Processing (NLP) and Transformer-based models to analyze the semantic meaning and contextual information contained in email subjects and message bodies. A pre-trained Transformer model can be fine-tuned to classify emails into relevant categories such as work-related, personal, promotional, spam, or informational. The system also predicts the priority of each email as High, Medium, or Low, helping users identify messages that require immediate attention.

By utilizing the self-attention mechanism of Transformer models, the system can understand contextual relationships between words more effectively than traditional rule-based and machine learning approaches. The model is evaluated using performance metrics such as accuracy, precision, recall, and F1-score.

The proposed system aims to reduce email overload, improve inbox organization, increase productivity, and minimize the possibility of missing important communications. Overall, this project demonstrates how Transformer-based Artificial Intelligence can provide an intelligent and efficient solution for automated email management and priority detection.

Overall, this project demonstrates the potential of Transformer-based Artificial Intelligence in developing intelligent email management systems. By combining advanced NLP techniques with automatic classification and priority prediction, the proposed system provides a more efficient, accurate, and context-aware approach to managing large volumes of email communication.

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CHAPTER 1: INTRODUCTION

1.1-Background Information

Email has become one of the most important communication tools for individuals, businesses, educational institutions, and organizations. Every day, users receive a large number of emails containing work requests, meeting notifications, advertisements, newsletters, personal messages, spam, and urgent communications. Managing this continuously growing volume of email manually can be difficult and time-consuming. Important messages may be overlooked, while users may spend unnecessary time reading emails that require little or no immediate attention. Therefore, intelligent systems for automatically classifying emails and identifying their priority have become increasingly important.

1.2-Project Objectives

The main objective of the project "AI-Based Email Classification and Priority Detection Using Transformers" is to develop an intelligent system that automatically classifies and prioritizes emails using Artificial Intelligence (AI), Natural Language Processing (NLP), and Transformer-based models. The system aims to analyze the subject and content of incoming emails and categorize them into relevant groups such as work-related, personal, promotional, spam, and informational emails. It also aims to detect the priority level of each email by classifying it as High, Medium, or Low based on its contextual meaning, urgency, deadlines, requests, and required actions. By using Transformer models, the system can understand the semantic relationships and context within email messages more effectively than traditional keyword-based methods.

1.3-Significance

The significance of the project "AI-Based Email Classification and Priority Detection Using Transformers" lies in its ability to improve the efficiency and intelligence of email management. With the increasing volume of emails received by individuals and organizations every day, manually sorting and identifying important messages can be difficult, time-consuming, and prone to errors. The proposed system helps overcome this problem by automatically classifying emails into meaningful categories and identifying their priority levels using Artificial Intelligence and Transformer-based Natural Language Processing techniques.

1.4-Scope

The scope of the project "AI-Based Email Classification and Priority Detection Using Transformers" focuses on developing an intelligent system capable of automatically analyzing, classifying, and prioritizing email messages using Artificial Intelligence, Natural Language Processing, and Transformer-based models. The system will primarily process email subjects and message bodies to understand the content and contextual meaning of each message. Based on this analysis, emails can be classified into predefined categories such as work-related, personal, promotional, spam, informational, or other relevant categories. The project includes data collection, text prepossessing, tokenization, model training, fine-tuning of a pre-trained Transformer model, prediction, and performance evaluation using metrics such as accuracy, precision, recall, and F1-score. The proposed system can be applied to personal and organizational email management environments to reduce information overload and improve productivity.

1.5-Methodology Overview

The methodology for the project "AI-Based Email Classification and Priority Detection Using Transformers" follows a systematic approach to develop an intelligent system for automatically analyzing, classifying, and prioritizing email messages. The process begins with the collection of a labeled email datasets containing email subjects, message bodies, categories, and priority information. The collected data is then prepossessed by removing unnecessary symbols, duplicate records, and irrelevant information, while preserving meaningful textual content. The email subject and body are combined and converted into a suitable format for analysis.

Methodology Overview

1. **Data Collection**
   Collect a labeled email data-set containing email subjects, message bodies, email categories, and priority levels.
2. **Data Prepossessing**
   Clean the collected email data by removing duplicate records, unnecessary symbols, irrelevant information, and missing values while retaining meaningful content.
3. **Feature Preparation**
   Combine important email fields such as the subject and message body to create meaningful input for the model.
4. **Text Tokenization**
   Convert the email text into tokens using the tokenism associated with the selected pre-trained Transformer model.

CHAPTER 2: PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Email has become an essential communication tool for individuals, educational institutions, businesses, and organizations. However, the increasing number of emails received every day creates a major challenge in managing and identifying important messages. Users often receive a large mixture of work-related emails, personal messages, promotional content, newsletters, spam, meeting notifications, and urgent requests. Manually reading and organizing these emails is time-consuming and can lead to information overload. As a result, important or urgent emails may be overlooked among less important messages, causing delayed responses, missed deadlines, and reduced productivity.

2.2 Evidence of the Problem

The problem of email overload is evident in the large number of messages received by individuals and organizations every day. Users often receive different types of emails, including work-related messages, meeting notifications, newsletters, promotional emails, spam, personal communication, and urgent requests. Because of this high volume, manually reading, sorting, and identifying important emails becomes difficult and time-consuming. Important messages can easily be hidden among less relevant emails, increasing the possibility of delayed responses, missed deadlines, and overlooked tasks.

2.3 Stakeholders

The stakeholders of the project **"AI-Based Email Classification and Priority Detection Using Transformers"** include all individuals and organizations that are directly or indirectly involved in, affected by, or benefit from the proposed system.

1. **Email Users** – Individual users are the primary stakeholders who benefit from automatic email classification and priority detection. The system helps them organize their inboxes, identify important messages, and reduce the time spent managing emails.
2. **Employees and Professionals** – Employees working in organizations receive a large number of work-related emails daily. They can benefit from the system by quickly identifying urgent tasks, meeting notifications, deadlines, and important communications.
3. **Organizations and Businesses** – Companies and institutions can use the system to improve communication efficiency, employee productivity, and email management.

CHAPTER 3: SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development and design process of the **AI-Based Email Classification and Priority Detection Using Transformers** system follows a structured approach to ensure that the proposed solution is efficient, accurate, and user-friendly. The process begins with identifying the problem of email overload and defining the functional requirements of the system. The main requirements include the ability to process email content, classify emails into predefined categories, and identify their priority levels.

The next stage involves collecting and preparing a suitable email data-set. The data-set contains email subjects, message bodies, category labels, and priority labels. The collected data is cleaned and prepossessed to remove duplicate records, unnecessary symbols, missing values, and irrelevant information. The subject and body of each email are then prepared as input text for the system..

3.2 Tools and Technologies Used

The development of the AI-Based Email Classification and Priority Detection Using Transformers system requires a combination of programming languages, machine learning frameworks, Natural Language Processing libraries, and development tools. Python is used as the primary programming language because of its simplicity and extensive support for Artificial Intelligence, Machine Learning, and data analysis. Libraries such as Pandas and NumPy are used for data collection, manipulation, cleaning, and prepossessing.

For Natural Language Processing and Transformer-based modeling, the Hugging Face Transformers library can be used to access pre-trained models and tokenizers. A suitable pre-trained Transformer model, such as BERT, can be fine-tuned using the email data-set to understand the contextual and semantic meaning of email content. The PyTorch or TensorFlow framework can be used for training and evaluating the deep learning model.

3.3 Solution Overview

The proposed solution for "AI-Based Email Classification and Priority Detection Using Transformers" is an intelligent email management system designed to automatically analyze the content of emails, classify them into appropriate categories, and determine their priority levels. The system uses Artificial Intelligence, Natural Language Processing, and a Transformer-based model to understand the contextual and semantic meaning of email messages. The main objective is to reduce the effort required for manual email organization and help users quickly identify important and urgent communications.

3.3 Engineering Standards Applied

The development of the "AI-Based Email Classification and Priority Detection Using Transformers" system follows appropriate software engineering and Artificial Intelligence development standards to ensure that the system is reliable, secure, maintainable, and efficient. A structured software development process is followed throughout the project, including requirement analysis, system design, implementation, testing, evaluation, and maintenance. The system is designed using modular components so that functions such as data prepossessing, email classification, priority detection, and result generation can be developed and maintained independently.

CHAPTER 4: RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The results of the AI-Based Email Classification and Priority Detection Using Transformers system are evaluated to determine how accurately and effectively the proposed model performs its two main tasks: email classification and priority detection. After training and fine-tuning the Transformer-based model, its performance is tested using a separate testing data-set that was not used during the training process. The predicted results are then compared with the actual labels to measure the correctness of the system.

The evaluation is performed using standard machine learning performance metrics such as accuracy, precision, recall, and F1-score. Accuracy measures the overall percentage of emails correctly classified by the system. Precision measures how many of the emails predicted as belonging to a particular category or priority level are actually correct. Recall measures the ability of the system to identify all relevant emails within a category, while the F1-score provides a balanced measurement of both precision and recall.

4.2 Challenges Encountered

During the development of the **AI-Based Email Classification and Priority Detection Using Transformers** system, several challenges were encountered. One of the major challenges was obtaining a suitable and properly labeled email data-set. Email classification and priority detection require data with accurate category and priority labels, but assigning priority levels such as High, Medium, and Low can be subjective because the importance of an email may vary depending on the user, sender, context, and situation.

Another challenge was the preprocessing of email data. Emails may contain unnecessary information such as HTML tags, URLs, signatures, special characters, duplicate content, and forwarded messages. Cleaning this data while preserving the meaningful context of the email was important for achieving better model performance.Handling imbalanced datasets was also a significant challenge. Some email categories or priority levels may contain a large number of examples, while urgent or high-priority emails may have fewer examples. This imbalance can cause the model to perform better on common categories and less effectively on underrepresented categories.

4.3 Possible Improvements

The proposed AI-Based Email Classification and Priority Detection Using Transformers system can be further improved in several ways. One possible improvement is the use of a larger and more diverse email dataset to improve the model's ability to understand different writing styles, topics, and communication patterns. Better data labeling and balancing of email categories and priority levels can also help improve the accuracy of the system, especially for underrepresented classes such as high-priority or urgent emails.

The system can also be improved by using more advanced Transformer models or comparing the performance of multiple models to identify the most suitable approach. Hyperparameter tuning, additional fine-tuning, and improved preprocessing techniques may further increase the accuracy and reliability of email classification and priority detection. Personalized learning can also be introduced so that the system gradually learns the preferences and email-handling behavior of individual users.

4.4 Recommendations

Based on the development and evaluation of the AI-Based Email Classification and Priority Detection Using Transformers system, several recommendations can be made to improve its effectiveness and practical application. It is recommended to use a large, diverse, and properly labeled email data-set to ensure that the system can accurately classify different types of emails and detect their priority levels. The data-set should include sufficient examples for each email category and priority level to reduce the problem of class imbalance and improve the detection of important or urgent emails.

It is also recommended to regularly fine-tune and update the Transformer model using new and relevant email data. This will help the system adapt to changing communication patterns, writing styles, and user requirements. Different Transformer models can be tested and compared to identify the model that provides the best balance between accuracy, processing speed, and computational requirements.

CHAPTER 5: REFLECTION ON LEARNING DEVELOPMENT

5.1 Key Learning Outcomes

The development of the AI-Based Email Classification and Priority Detection Using Transformers project provided several important learning outcomes in the areas of Artificial Intelligence, Natural Language Processing, Machine Learning, and software development. One of the main learning outcomes was understanding how Natural Language Processing techniques can be used to analyze and process unstructured text data such as email messages. The project also provided knowledge about text preprocessing, tokenization, data cleaning, and preparing textual data for machine learning models.

Another important learning outcome was gaining practical knowledge of Transformer-based models and their ability to understand the contextual and semantic meaning of text. The project helped in understanding how pre-trained Transformer models can be fine-tuned for specific tasks such as email classification and priority detection. It also provided experience in training, testing, and optimizing machine learning models.

Technical Skills:

The development of the AI-Based Email Classification and Priority Detection Using Transformers project helped develop and improve several technical skills. These include proficiency in Python programming for implementing the system and processing data. Skills in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) were developed through the analysis and classification of email text.

Problem-Solving:

The development of the AI-Based Email Classification and Priority Detection Using Transformers project enhanced problem-solving skills by requiring the identification, analysis, and resolution of various technical and practical challenges. The project involved finding an effective solution to the problem of email overload and the difficulty of manually identifying important and urgent messages. Different approaches were analyzed to determine how Artificial Intelligence, Natural Language Processing, and Transformer-based models could provide a more accurate and efficient solution.

Overall, the project improved the ability to analyze complex problems, identify suitable solutions, test different approaches, evaluate results, and make improvements based on the findings. These problem-solving skills are valuable for the development of future Artificial Intelligence, Machine Learning, and software engineering projects.

5.2 Challenges Encountered and Overcome

During the development of the AI-Based Email Classification and Priority Detection Using Transformers system, several technical and practical challenges were encountered and addressed. One of the major challenges was obtaining a suitable and properly labeled email data-set. Email data can contain incomplete, duplicate, irrelevant, or inconsistent information, making it difficult to use directly for model training. This challenge was overcome through data cleaning, prepossessing, removal of duplicate records, and proper organization of the datasets before training the model.

Another significant challenge was handling the unstructured nature of email content. Emails may contain HTML tags, special characters, signatures, URLs, forwarded messages, and unnecessary text. These elements can affect the performance of the model if they are not processed correctly. This issue was addressed by applying appropriate text prepossessing techniques to remove irrelevant information while preserving the meaningful context of the email.

5.3 Application of Engineering Standards

The development of the AI-Based Email Classification and Priority Detection Using Transformers system applies fundamental engineering standards and best practices throughout the project life cycle to ensure quality, reliability, security, and maintainability. A structured software engineering approach is followed, beginning with requirement analysis and continuing through system design, implementation, testing, evaluation, and improvement. The system is designed using modular components, including data collection, data prepossessing, Transformer-based classification, priority detection, and result presentation, making the application easier to develop, test, maintain, and extend.

5.4 Insights into the Industry

The development of the AI-Based Email Classification and Priority Detection Using Transformers project provides valuable insights into the growing role of Artificial Intelligence in the software and communication technology industry. Modern organizations increasingly depend on digital communication, resulting in large volumes of emails and messages that must be managed efficiently. This creates a growing demand for intelligent automation systems that can reduce manual work, organize information, and help users identify important communications quickly.

CHAPTER 6: CONCLUSION

The AI-Based Email Classification and Priority Detection Using Transformers project demonstrates how Artificial Intelligence, Natural Language Processing, and Transformer-based models can be used to improve the management of large volumes of email communication. The increasing number of emails received by individuals and organizations creates challenges such as information overload, time-consuming manual organization, and the possibility of overlooking important or urgent messages. The proposed system addresses these challenges by automatically analyzing email content, classifying messages into appropriate categories, and assigning priority levels such as High, Medium, or Low.

By using Transformer-based models, the system can understand the contextual and semantic meaning of email subjects and message bodies more effectively than traditional rule-based or keyword-based approaches. This enables the system to identify the category and importance of an email based on its overall meaning, deadlines, requests, and required actions. The use of machine learning evaluation metrics such as accuracy, precision, recall, and F1-score helps measure the performance and reliability of the system.

The project also provides valuable learning and practical experience in areas such as Artificial Intelligence, Natural Language Processing, data prepossessing, Transformer models, model training, evaluation, and software development. Several challenges, including data quality, class imbalance, contextual understanding, computational requirements, and privacy concerns, were identified and addressed during the development process.

Overall, the proposed system provides an intelligent and efficient solution for email classification and priority detection. It has the potential to reduce information overload, improve inbox organization, increase productivity, and help users respond more quickly to important communications. With further improvements such as personalized learning, multilingual support, additional contextual features, real-time email integration, and enhanced security mechanisms, the system can be developed into a more advanced and practical solution for personal and organizational email management.

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Appendix:

```
# ==========================================================
# APPENDIX
# AI-Based Email Classification and Priority Detection
# Using Transformer
# ==========================================================

# -----------------------------
# Appendix A: Import Libraries
# -----------------------------

from transformers import pipeline

# Load Transformer-based text classification model
classifier = pipeline("text-classification")


# -----------------------------
# Appendix B: Sample Emails
# -----------------------------

emails = [
    "Urgent! Please submit the project report before 5 PM today.",
    "Your bank account has been blocked. Take immediate action.",
    "Get 50% discount on all products. Shop now!",
    "Team meeting is scheduled for tomorrow at 10 AM.",
    "Congratulations! You have received a special offer.",
    "The project server is down. Immediate support is required."
]


# -----------------------------
# Appendix C: Email Classification
# -----------------------------

for email in emails:

    result = classifier(email)

    print("Email:", email)
    print("Model Result:", result)
    print("-" * 60)


# -----------------------------
# Appendix D: Priority Detection
# -----------------------------
```

```
def detect_priority(email):

    high_words = [
        "urgent",
        "immediate",
        "asap",
        "deadline",
        "blocked",
        "emergency",
        "critical"
    ]

    medium_words = [
        "meeting",
        "reminder",
        "notification",
        "schedule",
        "report"
    ]

    email = email.lower()

    for word in high_words:
        if word in email:
            return "High"

    for word in medium_words:
        if word in email:
            return "Medium"

    return "Low"


# -----------------------------
# Appendix E: Final Prediction
# -----------------------------

for email in emails:

    priority = detect_priority(email)

    print("Email:", email)
    print("Priority:", priority)
    print("=" * 60)


# -----------------------------
# Appendix F: Complete System
# -----------------------------
```

```
def email_analysis(email):

    # Transformer classification
    classification = classifier(email)

    # Priority detection
    priority = detect_priority(email)

    return classification, priority


# Test email
email = """
Urgent! The project server is not responding.
Please fix the problem immediately.
"""

classification, priority = email_analysis(email)

print("\nFinal Email Analysis")
print("--------------------")
print("Email:", email)
print("Classification:", classification)
print("Priority:", priority)
```